

Scientific Words of the Day

30 root puzzles from physics to environmental science

Physics first, Earth systems next

prefix + root + suffix → physical model

How to use one pair of slides each day

1. Read the root cards aloud.
2. Build and justify a possible word.
3. Sketch the process or system boundary.
4. Turn the page and read the word in context.
5. Answer the final check question.

Recurring questions

What moves? In what form? Across which boundary? What changes? What remains unchanged?

Root key

L = Latin origin

G = Greek origin

Teacher note

The root gloss is a memory prompt. The scientific context supplies the precise modern meaning. Many science words combine Greek and Latin elements.

Day 1 | Build the word

LIGHT, RADIATION AND TRANSFER

back + bend + process

re-
back L

flect
bend L

-ion
act or process L

Phenomenon clue A wave reaches a boundary and returns to the original medium.

Build the scientific word.

Then say what you think it means before turning the page.

reflection

Root image: the process of bending back

In context Light undergoes **reflection** when it returns from a mirror or water surface.

Physics

For a plane mirror, the angles are measured from the normal: $\theta_i = \theta_r$.

Environmental connection

Ice and clouds reflect sunlight. Their reflectivity contributes to Earth's albedo.

Word family: reflect, reflection, reflector, reflective

Check: What changes in reflection: the wave speed, its direction, or both?

Day 2 | Build the word

LIGHT, RADIATION AND TRANSFER

back + break + process

re-
back L

fract
break L

-ion
act or process L

Phenomenon clue A wave crosses into a new medium, changes speed and usually changes direction.

Build the scientific word.

Then say what you think it means before turning the page.

refraction

Root image: a bending or “breaking” of the path

In context Atmospheric **refraction** can make the Sun visible just before geometric sunrise.

Physics

The refractive index is $n = c/v$. A change in wave speed produces a change in direction.

Environmental connection

Layers of warm and cool air refract light differently, producing mirages.

Word family: refract, refraction, refractive, refractor

Check: Why does a ray entering along the normal change speed without changing direction?

Day 3 | Build the word

LIGHT, RADIATION AND TRANSFER

across + send + process

trans-

across L

miss

send L

-ion

act or process L

Phenomenon clue Some of the incident wave passes through the material or boundary.

Build the scientific word.

Then say what you think it means before turning the page.

transmission

Root image: the process of sending across

In context Glass allows high **transmission** of visible light.

Physics

A transmission fraction can be written

$$T = I_{\text{trans}} / I_{\text{incident}}.$$

Environmental connection

The atmosphere has transmission windows through which visible light and some radio waves pass.

Word family: transmit, transmission, transmitter, transmissive

Check: If 70% is transmitted, what fraction remains for reflection and absorption?

Day 4 | Build the word

LIGHT, RADIATION AND TRANSFER

away or into + swallow + process

ab-

away or into L

sorpt

swallow up L

-ion

act or process L

Phenomenon clue Energy carried by a wave is taken into a material and changes its internal energy.

Build the scientific word.

Then say what you think it means before turning the page.

absorption

Root image: the process of swallowing up

In context Dark surfaces show strong **absorption** of visible radiation.

Physics

Absorbed radiation transfers energy to matter, often raising its internal energy.

Environmental connection

Greenhouse gases absorb selected infrared wavelengths; oceans absorb most of Earth's excess heat.

Word family: absorb, absorption, absorptive, absorber

Check: How is absorption different from a wave merely disappearing from view?

Day 5 | Build the word

LIGHT, RADIATION AND TRANSFER

out + send + process

e- / ex-
out L

miss
send L

-ion
act or process L

Phenomenon clue Radiation or matter leaves a source and enters its surroundings.

Build the scientific word.

Then say what you think it means before turning the page.

emission

Root image: the process of sending out

In context Every warm body produces thermal **emission**.

Physics

A grey body's radiated power is $P = \varepsilon\sigma AT^4$.

Environmental connection

Earth emits infrared radiation; vehicles also emit gases and particles. The same word can name energy or matter leaving a system.

Word family: emit, emission, emitter, emissivity

Check: In each use of “emission”, what crosses the system boundary?

Day 6 | Build the word

LIGHT, RADIATION AND TRANSFER

together + lead + process

con-
together **L**

duct
lead **L**

-ion
act or process **L**

Phenomenon clue Energy passes through matter by interactions between neighbouring particles.

Build the scientific word.

Then say what you think it means before turning the page.

conduction

Root image: the process of leading together

In context A metal spoon warms by thermal **conduction**.

Physics

For a simple slab, the transfer rate is $P = kA\Delta T/L$.

Environmental connection

Conduction through walls affects building energy demand; soil conduction affects permafrost thaw.

Word family: conduct, conduction, conductor, conductivity

Check: Which material property in the equation measures how readily energy is conducted?

Day 7 | Build the word

LIGHT, RADIATION AND TRANSFER

together + carry + process

con-
together L

vect
carry L

-ion
act or process L

Phenomenon clue A moving fluid carries internal energy from one region to another.

Build the scientific word.

Then say what you think it means before turning the page.

convection

Root image: the process of carrying together

In context Warm air rises and cooler air replaces it in a **convection** current.

Physics

Convection combines bulk fluid motion with energy transfer. It requires a liquid or gas.

Environmental connection

Atmospheric and oceanic convection redistribute heat, moisture and dissolved substances.

Word family: convect, convection, convective; compare advection

Check: What is transported in convection besides energy?

Day 8 | Build the word

LIGHT, RADIATION AND TRANSFER

ray + process

radi-

ray or spoke L

-ation

act or process L

Phenomenon clue Energy travels from a source as waves and needs no material medium.

Build the scientific word.

Then say what you think it means before turning the page.

radiation

Root image: energy travelling as rays

In context The Sun transfers energy through space by electromagnetic **radiation**.

Physics

Electromagnetic waves in vacuum obey $c = f\lambda$.

Environmental connection

Incoming solar radiation and outgoing infrared radiation form Earth's planetary energy budget.

Word family: radiate, radiation, radiator, irradiance

Check: Why can radiation cross space while conduction and convection cannot?

Day 9 | Build the word

LIGHT, RADIATION AND TRANSFER

island + process

insul(a)

island L

-ation

act or process L

Phenomenon clue A material separates a region from energy transfer, rather like making it an island.

Build the scientific word.

Then say what you think it means before turning the page.

insulation

Root image: the process of making into an island

In context Foam provides thermal **insulation** by reducing energy transfer.

Physics

Insulation lowers the transfer rate; it does not prevent energy transfer perfectly.

Environmental connection

Building insulation reduces fuel demand. Snow also insulates soil and organisms beneath it.

Word family: insulate, insulation, insulator, insulating

Check: Which transfer mechanisms can a layer of insulation reduce?

Day 10 | Build the word

LIGHT, RADIATION AND TRANSFER

between + do + process

inter-

between L

act

do or drive L

-ion

act or process L

Phenomenon clue Two parts of a system affect one another through a force, collision or exchange.

Build the scientific word.

Then say what you think it means before turning the page.

interaction

Root image: an action between things

In context A force describes an **interaction** between bodies.

Physics

Interactions occur in pairs: if A exerts a force on B, B exerts an equal and opposite force on A.

Environmental connection

Environmental systems contain linked physical, chemical and biological interactions.

Word family: interact, interaction, interactive, actor

Check: Why is a force better represented between two bodies than as a possession of one body?

Day 11 | Build the word

MOTION, FORCES AND SYSTEMS

toward or more + swift + process

ac-

toward; intensifies L

celer

swift L

-ation

act or process L

Phenomenon clue The velocity of an object changes with time.

Build the scientific word.

Then say what you think it means before turning the page.

acceleration

Root image: the process of becoming swifter

In context A falling raindrop undergoes downward **acceleration** until drag becomes large enough to balance its weight.

Physics

Average acceleration is $a = \Delta v / \Delta t$. Direction may change even when speed does not.

Environmental connection

Wind acceleration and changing currents influence the transport of heat, moisture and pollutants.

Word family: accelerate, acceleration, accelerator

Check: Can an object accelerate while moving at constant speed? Give an example.

Day 12 | Build the word

MOTION, FORCES AND SYSTEMS

heavy + process

gravi-

heavy L

-ation

act or process L

Phenomenon clue Every mass interacts attractively with every other mass.

Build the scientific word.

Then say what you think it means before turning the page.

gravitation

Root image: the process associated with heaviness

In context Earth's **gravitation** gives falling objects their weight.

Physics

For two point masses, $F = GMm/r^2$. The field is an interaction with mass.

Environmental connection

Gravity drives sediment settling, river flow, atmospheric pressure gradients and ocean tides.

Word family: gravity, gravitation, gravitational, gravimeter

Check: Which variable in the inverse-square law describes separation rather than either object?

Day 13 | Build the word

MOTION, FORCES AND SYSTEMS

toward + draw + process

at-
toward L

tract
draw or pull L

-ion
act or process L

Phenomenon clue An interaction tends to draw two bodies or particles towards one another.

Build the scientific word.

Then say what you think it means before turning the page.

attraction

Root image: the process of drawing towards

In context Unlike electric charges experience electrostatic **attraction**.

Physics

Attraction describes the direction of the force; it does not identify the interaction responsible.

Environmental connection

Attractive forces help aerosol particles cluster and affect how contaminants attach to surfaces.

Word family: attract, attraction, attractive, tractor

Check: What additional information is needed before “attraction” becomes a physical explanation?

Day 14 | Build the word

MOTION, FORCES AND SYSTEMS

together + press + process

com-
together L

press
press L

-ion
act or process L

Phenomenon clue A force reduces the volume of a material, most noticeably for a gas.

Build the scientific word.

Then say what you think it means before turning the page.

compression

Root image: the process of pressing together

In context Rapid **compression** of a gas usually raises its temperature.

Physics

Work done on a gas transfers energy into it. At fixed temperature, pV is approximately constant for an ideal gas.

Environmental connection

Descending air is compressed and warms, influencing cloud formation and local weather.

Word family: compress, compression, compressor, compressible

Check: Why does compression require an energy-transfer account as well as a volume account?

Day 15 | Build the word

MOTION, FORCES AND SYSTEMS

out + spread + process

ex-

out **L**

pans

spread **L**

-ion

act or process **L**

Phenomenon clue A material occupies more volume; a rising gas may cool as it does work.

Build the scientific word.

Then say what you think it means before turning the page.

expansion

Root image: the process of spreading out

In context A gas cools during adiabatic **expansion** because it does work on its surroundings.

Physics

Thermal expansion also changes the dimensions of solids and liquids as temperature changes.

Environmental connection

Rising air expands and cools; water vapour may then condense to form clouds.

Word family: expand, expansion, expansive, expansivity

Check: What system does the expanding gas do work on?

Day 16 | Build the word

MOTION, FORCES AND SYSTEMS

swing + process

oscill-

swing L

-ation

act or process L

Phenomenon clue A system repeatedly moves or varies about an equilibrium value.

Build the scientific word.

Then say what you think it means before turning the page.

oscillation

Root image: the process of swinging

In context A pendulum's displacement undergoes an **oscillation** about equilibrium.

Physics

Period and frequency are reciprocal: $T = 1/f$.
Energy shifts between stores during each cycle.

Environmental connection

Ocean-atmosphere patterns such as El Niño involve large-scale, irregular oscillations.

Word family: oscillate, oscillation, oscillator, oscillatory

Check: What defines the equilibrium value about which a system oscillates?

Day 17 | Build the word

MOTION, FORCES AND SYSTEMS

equal + balance + state

equi-

equal L

libr

balance L

-ium

state or result L

Phenomenon clue Opposing processes continue, yet their rates are equal and the macroscopic state is steady.

Build the scientific word.

Then say what you think it means before turning the page.

equilibrium

Root image: a state of equal balance

In context At thermal **equilibrium**, bodies in contact have no net energy transfer between them.

Physics

Equilibrium is often dynamic: microscopic exchanges continue while opposing rates balance.

Environmental connection

Planetary radiative equilibrium requires average incoming and outgoing energy fluxes to be equal.

Word family: equilibrate, equilibrium, equilibrant

Check: Why does equilibrium mean zero net change rather than zero activity?

Day 18 | Build the word

MOTION, FORCES AND SYSTEMS

together + keep + process

con-
together L

serv
keep L

-ation
act or process L

Phenomenon clue A total remains unchanged even while it is redistributed or transformed.

Build the scientific word.

Then say what you think it means before turning the page.

conservation

Root image: the process of keeping together

In context Energy **conservation** requires the total energy of a closed system to remain constant.

Physics

A conservation law is a bookkeeping constraint:
 $\Delta E_{\text{total}} = 0$ for an isolated system.

Environmental connection

Environmental conservation means protection and careful management. “Conserve energy” concerns use and quality, since the amount itself is conserved.

Word family: conserve, conservation, conservative, conserved

Check: How do the physics and environmental meanings of conservation differ?

Day 19 | Build the word

MOTION, FORCES AND SYSTEMS

apart + scatter + process

dis-
apart L

sipat
scatter L

-ion
act or process L

Phenomenon clue Concentrated energy spreads into less useful thermal stores.

Build the scientific word.

Then say what you think it means before turning the page.

dissipation

Root image: the process of scattering apart

In context Friction causes mechanical energy **dissipation** into thermal energy stores.

Physics

Energy remains conserved while becoming more dispersed and less available for useful work.

Environmental connection

Engines, electrical systems and cities dissipate energy as waste heat; efficiency limits fuel demand and environmental impact.

Word family: dissipate, dissipation, dissipative

Check: How can energy be both conserved and dissipated?

Day 20 | Build the word

MOTION, FORCES AND SYSTEMS

across or change + shape + process

trans-

across or change L

form

shape L

-ation

act or process L

Phenomenon clue A quantity or system changes form while accounting remains continuous.

Build the scientific word.

Then say what you think it means before turning the page.

transformation

Root image: the process of changing shape or form

In context A motor produces an energy **transformation** from electrical input to kinetic and thermal stores.

Physics

A transformation changes representation or form; a transfer crosses a system boundary.

Environmental connection

Photosynthesis transforms radiant energy into chemical energy stored in biomass.

Word family: transform, transformation, transformer, transformative

Check: Is heating a transformation, a transfer, or potentially both? Define the system first.

Day 21 | Build the word

ENVIRONMENTAL SYSTEMS

defile or make unclean + result

pollut-
defile or soil **L**

-ion
act or result **L**

Phenomenon clue Matter or energy enters an environment at a level that causes harm.

Build the scientific word.

Then say what you think it means before turning the page.

pollution

Root image: the result of making an environment unclean

In context Air **pollution** includes harmful gases and suspended particles.

Physics

Physics describes sources, diffusion, flow, settling and exposure; harm also depends on chemistry and biology.

Environmental connection

Pollution may involve matter, noise, light or thermal energy. It is defined partly by its damaging effects.

Word family: pollute, pollution, pollutant, polluting

Check: Why is the presence of a substance alone insufficient to establish pollution?

Day 22 | Build the word

ENVIRONMENTAL SYSTEMS

down + step + process

de-
down L

grad
step L

-ation
act or process L

Phenomenon clue A system moves to a lower condition, quality or capacity.

Build the scientific word.

Then say what you think it means before turning the page.

degradation

Root image: the process of stepping down

In context Energy **degradation** describes the loss of its capacity to produce useful work.

Physics

The total energy remains conserved while dissipation lowers its usefulness.

Environmental connection

Land degradation reduces soil structure, fertility, biodiversity or productive capacity.

Word family: degrade, degradation, degradable, biodegradable

Check: Which measurable indicators would show that an ecosystem has degraded?

ENVIRONMENTAL SYSTEMS

vapour + sphere

atmos

vapour **G**

sphere

globe **G**

Phenomenon clue A gravitationally bound layer of gases surrounds a planet.

Build the scientific word.

Then say what you think it means before turning the page.

atmosphere

Root image: the globe or region of vapour

In context Earth's **atmosphere** is a compressible fluid held by gravity.

Physics

Pressure, density, convection, radiation and phase changes govern its behaviour.

Environmental connection

The atmosphere transports energy and water, shields life and exchanges matter with oceans, land and organisms.

Word family: atmosphere, atmospheric; compare hydrosphere and biosphere

Check: Why is the atmosphere best modelled as both a layer and a dynamic system?

Day 24 | Build the word

ENVIRONMENTAL SYSTEMS

human + produced by

anthropo-
human **G**

-genic
produced by **G**

Phenomenon clue A change, substance or forcing originates in human activity.

Build the scientific word.

Then say what you think it means before turning the page.

anthropogenic

Root image: produced by humans

In context The observed rise in atmospheric carbon dioxide has a large **anthropogenic** component.

Physics

A causal claim requires a mechanism, measurements and comparison with natural drivers.

Environmental connection

Anthropogenic forcing includes greenhouse-gas emissions, aerosols and land-use change.

Word family: anthropogenic, anthropogenicity; compare biogenic and geogenic

Check: Why does “anthropogenic” identify an origin rather than automatically assign blame?

Day 25 | Build the word

ENVIRONMENTAL SYSTEMS

life + difference + state

bio-
life **G**

divers
different **L**

-ity
state or quality **L**

Phenomenon clue Variation exists among genes, species and ecosystems.

Build the scientific word.

Then say what you think it means before turning the page.

biodiversity

Root image: the state of varied life

In context A coral reef may contain high **biodiversity** across several levels of organisation.

Physics

Biodiversity is measured through richness, evenness and related indices; no single number captures every aspect.

Environmental connection

Diverse systems may possess more response pathways when conditions change, though diversity alone does not guarantee resilience.

Word family: biology, biological, diversity, biodiverse

Check: What information is lost if biodiversity is reported only as a species count?

Day 26 | Build the word

ENVIRONMENTAL SYSTEMS

life + heap up + process

bio-
life **G**

accumul
heap up **L**

-ation
act or process **L**

Phenomenon clue An organism takes in a substance faster than it can remove or break it down.

Build the scientific word.

Then say what you think it means before turning the page.

bioaccumulation

Root image: the process of heaping up in life

In context Mercury can undergo **bioaccumulation** within an individual organism over time.

Physics

A mass balance compares input rate with metabolism and excretion. Accumulation occurs when net input is positive.

Environmental connection

Persistent, fat-soluble substances are especially likely to accumulate in tissues.

Word family: accumulate, accumulation, bioaccumulate, bioaccumulation

Check: What two rates must be compared to predict whether concentration will rise?

Day 27 | Build the word

ENVIRONMENTAL SYSTEMS

life + make greater + process

bio-
life **G**

magn
great **L**

fic
make **L**

-ation
process **L**

Phenomenon clue The concentration of a persistent contaminant increases at higher trophic levels.

Build the scientific word.

Then say what you think it means before turning the page.

biomagnification

Root image: the process of making greater through life

In context A persistent pesticide may show **biomagnification** from prey to predator.

Physics

A predator consumes contaminant stored in many prey organisms, producing a trophic concentration increase.

Environmental connection

Top predators may receive the largest dose even when environmental concentrations are low.

Word family: magnify, magnification, biomagnify, biomagnification

Check: How does biomagnification differ from bioaccumulation?

Day 28 | Build the word

ENVIRONMENTAL SYSTEMS

set apart + process

sequestr-

set apart L

-ation

act or process L

Phenomenon clue Carbon is removed from active circulation and stored in a reservoir.

Build the scientific word.

Then say what you think it means before turning the page.

sequestration

Root image: the process of setting apart

In context Carbon **sequestration** stores carbon in biomass, soils, oceans or geological formations.

Physics

A useful account specifies the reservoir, transfer rate, storage capacity and residence time.

Environmental connection

Storage contributes to mitigation only while carbon remains isolated from the atmosphere for a relevant timescale.

Word family: sequester, sequestration, sequestered

Check: Why is the amount stored insufficient without a timescale?

Day 29 | Build the word

ENVIRONMENTAL SYSTEMS

well + nourishment + process

eu-

well or abundant **G**

troph

nourishment **G**

-ication

process of making **L**

Phenomenon clue Excess nutrients stimulate growth and can eventually remove dissolved oxygen from water.

Build the scientific word.

Then say what you think it means before turning the page.

eutrophication

Root image: the process of becoming richly nourished

In context Fertiliser runoff can cause lake **eutrophication**.

Physics

Nutrients increase algal growth; decomposition then raises biological oxygen demand and lowers dissolved oxygen.

Environmental connection

A resource that initially increases production can destabilise the wider aquatic system.

Word family: trophic, eutrophic, eutrophication, oligotrophic

Check: Why can greater nutrient supply ultimately reduce animal life in the water?

Day 30 | Build the word

ENVIRONMENTAL SYSTEMS

hold up + capable + state

sus- / sub-
up or from below L

tain
hold L

-able
capable of L

-ity
state or quality L

Phenomenon clue A system can continue without exhausting the stocks and processes that support it.

Build the scientific word.

Then say what you think it means before turning the page.

sustainability

Root image: the capacity to keep holding up or enduring

In context Environmental **sustainability** asks whether a practice can persist within physical and ecological limits.

Physics

The question requires stocks, flows, efficiencies, feedbacks, boundaries and timescales.

Environmental connection

A process is sustainable only relative to stated environmental, social and temporal conditions.

Word family: sustain, sustainable, sustainability, sustained

Check: What would have to remain available or regenerate for this system to continue?

From word to world

root $\longrightarrow$ action $\longrightarrow$ model $\longrightarrow$ system

What moves? In what form? Across which boundary?

What changes? What accumulates? What remains unchanged?
